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DC/DC converter, electronic module, and electronic apparatus

Abstract

A DC/DC converter includes: multiple channels each including a switching circuit and an inductor; an output line having one end connected to the other end of the inductors of the multiple channels and the other end connected to a load; and a controller. The multiple channels include first to third channels, and the inductors of the first and second channels have a larger inductance value than the inductor of the third channel. When current flowing in the load is first current, the controller controls the switching circuits so that only the first channel is activated; and when the current flowing in the load is second current that is larger than the first current, the controller controls the switching circuits so that the first and second channels are operated in-phase and the third channel is operated in a phase different from the phase of the first and second channels.

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Background/Summary

BACKGROUND

Field

(1) The present disclosure relates to a DC/DC converter, an electronic module, and an electronic apparatus.

Description of the Related Art

(2) Japanese Patent Application Laid-Open No. 2014-226026 discloses a technology to improve efficiency in a light-load state of a multiphase DC/DC converter. In the technology disclosed in Japanese Patent Application Laid-Open No. 2014-226026, the inductance of an inductor of a single channel is set to a different value from inductances of inductors of the remaining channels so that high efficiency can be obtained in the lightest-load state where only the single channel is active.

(3) In the technology disclosed in Japanese Patent Application Laid-Open No. 2014-226026, however, there is a problem of increased ripple noise in the output voltage in a state where several channels of the multiphase DC/DC converter are activated and a large load current flows in the load.

SUMMARY

(4) Various embodiments of the present disclosure have been made in view of the above problem

and provide a DC/DC converter that can reduce ripple noise in the output voltage regardless of the level of load current

(5) According to one embodiment of the present disclosure, there is provided a DC/DC converter including: an input line supplied with a DC voltage; a plurality of channels each including a switching circuit and an inductor, one end of the inductor being connected to the input line via the switching circuit; an output line, one end of the output line being connected to the other end of the inductors of the plurality of channels, and the other end of the output line being connected to a load; and a controller that controls the switching circuits of the plurality of channels, wherein the plurality of channels include at least a first channel, a second channel, and a third channel, wherein the inductors of the first and second channels have a larger inductance value than the inductor of the third channel, wherein when current flowing in the load is first current, the controller controls the switching circuits of the first to third channels so that only the first channel is activated and the second and third channels are deactivated, and wherein when the current flowing in the load is second current that is larger than the first current, the controller controls the switching circuits of the first to third channels so that the first and second channels are operated in-phase and the third channel is operated in a phase different from the phase of the first and second channels.

(6) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A is a circuit diagram illustrating a circuit of a DC/DC converter according to a first embodiment of the present disclosure.

(2) FIG. **1**B is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the first embodiment of the present disclosure.

(3) FIG. **1**C is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the first embodiment of the present disclosure.

(4) FIG. **1**D is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the first embodiment of the present disclosure.

(5) FIG. **2** is a circuit diagram illustrating a circuit of a DC/DC converter according to a second embodiment of the present disclosure.

(6) FIG. **3**A is a circuit diagram illustrating a circuit of a DC/DC converter according to a comparative form.

(7) FIG. **3**B is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the comparative form.

(8) FIG. **3**C is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the comparative form.

(9) FIG. **3**D is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the comparative form.

(10) FIG. **3**E is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the comparative form.

(11) FIG. **3**F is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the comparative form.

(12) FIG. **4**A is a circuit diagram illustrating a circuit of a DC/DC converter according to a third embodiment of the present disclosure.

(13) FIG. **4**B is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the third embodiment of the present disclosure.

(14) FIG. **4**C is a diagram illustrating an example of an operation waveform of the DC/DC

converter according to the third embodiment of the present disclosure.
(15) FIG. **4**D is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the third embodiment of the present disclosure.
(16) FIG. **4**E is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the third embodiment of the present disclosure.
(17) FIG. **4**F is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the third embodiment of the present disclosure.
(18) FIG. **5**A is a diagram illustrating an example of an operation waveform of a DC/DC converter according to a fourth embodiment of the present disclosure.
(19) FIG. **5**B is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the fourth embodiment of the present disclosure.
(20) FIG. **5**C is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the fourth embodiment of the present disclosure.
(21) FIG. **5**D is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the fourth embodiment of the present disclosure.
(22) FIG. **5**E is a diagram illustrating an example of an operation waveform of the DC/DC converter according to the fourth embodiment of the present disclosure.
(23) FIG. **6**A is a sectional view illustrating a digital camera that is an imaging apparatus as one example of an electronic apparatus according to a fifth embodiment of the present disclosure.
(24) FIG. **6**B is a perspective view illustrating a processing module in the digital camera that is the imaging apparatus as one example of the electronic apparatus according to the fifth embodiment of the present disclosure.
(25) FIG. **6**C is a sectional view illustrating the processing module in the digital camera that is the imaging apparatus as one example of the electronic apparatus according to the fifth embodiment of the present disclosure.
(26) FIG. **7**A is a circuit diagram illustrating a circuit of a general multiphase DC/DC converter.
(27) FIG. **7**B is a circuit diagram illustrating the circuit of the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026.
(28) FIG. **7**C is a diagram illustrating an operation waveform of the general multiphase DC/DC converter when only a single channel is activated.
(29) FIG. **7**D is a diagram illustrating an operation waveform of the general multiphase DC/DC converter when only a single channel is activated.
(30) FIG. **7**E is a diagram illustrating an operation waveform of the general multiphase DC/DC converter when only a single channel is activated.
(31) FIG. **7**F is a diagram illustrating an operation waveform of the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 when only a single channel is activated.
(32) FIG. **7**G is a diagram illustrating an operation waveform of the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 when only a single channel is activated.
(33) FIG. **7**H is a diagram illustrating an operation waveform of the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 when only a single channel is activated.
(34) FIG. **7**I is a diagram illustrating an operation waveform of a general multiphase DC/DC converter when multiple channels are activated.
(35) FIG. **7**J is a diagram illustrating an operation waveform of the general multiphase DC/DC converter when multiple channels are activated.
(36) FIG. **7**K is a diagram illustrating an operation waveform of the general multiphase DC/DC converter when multiple channels are activated.
(37) FIG. **7**L is a diagram illustrating an operation waveform of the multiphase DC/DC converter

disclosed in Japanese Patent Application Laid-Open No. 2014-226026 when multiple channels are activated.
(38) FIG. 7M is a diagram illustrating an operation waveform of the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 when multiple channels are activated.
(39) FIG. 7N is a diagram illustrating an operation waveform of the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 when multiple channels are activated.
DESCRIPTION OF THE EMBODIMENTS
(40) [Reference Technology]
(41) Electronic modules mounted on electronic apparatuses each have a printed wiring board, a semiconductor apparatus implemented on the printed wiring board, and a power source circuit that supplies power to the semiconductor apparatus. In recent years, an increase in the amount of data processing per unit time has caused an increase in the amount of current required for the semiconductor apparatus to operate. Thus, DC/DC converters that do not cause a significant increase in the loss even with an increased amount of current are utilized as the power source circuit. While DC/DC converters have high efficiency, ripple noise is superimposed on the output voltage due to switching operations to control the levels of the output current and the output voltage. Advancement of semiconductor technologies and an increased demand for lower power consumption have led to continuous decrease in operation voltages of semiconductor apparatuses. Due to such a decrease in voltages, the amount of tolerant noise that is set for avoiding a malfunction of semiconductor apparatuses has also decreased, and DC/DC converters are thus required not only to increase efficiency but also to reduce ripple noise.
(42) Among semiconductor apparatuses, in a central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), or the like, the operation current decreases to a value close to zero in a standby state, and in contrast, the operation current significantly increases in accordance with an amount of data processing in an operating state. That is, the output current (load current) of a DC/DC converter will significantly vary within a range from several mA to several A. To address such a large dynamic range of load current, the multiphase DC/DC converter as disclosed in Japanese Patent Application Laid-Open No. 2014-226026 formed of a plurality of DC/DC converters connected in parallel is utilized. Such a multiphase DC/DC converter addresses the increase and decrease in the amount of current by activating only one DC/DC converter when the load such as a CPU or the like is in a standby state and activating all the DC/DC converters in a full operating state. In a standby state of semiconductor apparatuses, since only one DC/DC converter is activated, there is an advantage that the switching loss can be suppressed low. In addition, in a full operating state where all the DC/DC converters are activated, there is an advantageous effect that the total ripple noise is reduced by activating respective converters with shifted phases to cancel respective ripple noise to each other.
(43) When semiconductor apparatuses are in a standby state or the like, however, there is a problem of larger ripple noise than in a full operating state, because the above cancelling effect is not obtained when only a single DC/DC converter is activated. Accordingly, Japanese Patent Application Laid-Open No. 2014-226026 discloses a technology that can reduce ripple noise even when only a single DC/DC converter is activated, such as in a standby state. Specifically, in Japanese Patent Application Laid-Open No. 2014-226026, only the inductance value of the output inductor component of a DC/DC converter which is activated in a standby state of semiconductor apparatuses is set to a larger value, thereby the amplitude of current occurring due to a switching operation is reduced, and the ripple noise is reduced.
(44) As reference technologies, a general multiphase DC/DC converter and the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 will be described with reference to FIG. 7A to FIG. 7H.

(45) FIG. **7**A is a circuit diagram illustrating an example of a circuit of a general multiphase DC/DC converter. The example illustrated in FIG. **7**A represents a two-phase circuit. As illustrated in FIG. **7**A, a DC/DC converter **60**A that is a general multiphase DC/DC converter has switching circuits **602**A, **603**A, inductors **604**A, **605**A, a capacitor **606**A, and a controller **608**A. Further, the DC/DC converter **60**A has an input line **621**A and an output line **622**A.
(46) The input line **621**A is a wiring supplied with a direct current (DC) voltage from an input voltage source **601**A. The input voltage source **601**A is connected to one end of the input line **621**A. The input voltage source **601**A supplies a DC voltage to the input line **621**A. The input voltage source **601**A is not particularly limited and may be a power supply that supplies a DC voltage converted from alternating current (AC) power supplied from a commercial power supply or may be a battery, for example. The switching circuits **602**A, **603**A are connected in parallel to the other end of the input line **621**A via wirings.
(47) The switching circuits **602**A, **603**A are each formed of a complementary metal oxide semiconductor (CMOS) inverter having a P-type metal oxide semiconductor (MOS) transistor Tr**1** and an N-type MOS transistor Tr**2**. In each of the switching circuits **602**A, **603**A, the source of the P-type MOS transistor Tr**1** is connected to the other end of the input line **621**A via a wiring. Further, the source of the N-type MOS transistor Tr**2** is connected to a reference potential **609**A, which is a ground potential, via a wiring. Furthermore, the gate of the P-type MOS transistor Tr**1** and the gate of the N-type MOS transistor Tr**2** are connected to the controller **608**A via wirings. Further, the drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **602**A are connected to each other and connected as the output terminal to one end of the inductor **604**A via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **603**A are connected to each other and connected as the output terminal to one end of the inductor **605**A via a wiring.
(48) The controller **608**A is a control unit that controls switching operations of the switching circuits **602**A, **603**A to switch on and off the switching circuits **602**A, **603**A. The controller **608**A controls the switching operation by controlling the voltages supplied to the gates of the P-type MOS transistors Tr**1** and the gates of the N-type MOS transistors Tr**2** of the switching circuits **602**A, **603**A.
(49) The inductors **604**A, **605**A are inductors having the same inductance value L each other. The other end of the inductor **604**A and the other end of the inductor **605**A are connected to each other via wirings and connected to one end of the output line **622**A. In FIG. **7**A, current **610**A illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **604**A and one end of the output line **622**A. Current **611**A illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **605**A and one end of the output line **622**A.
(50) The output line **622**A is a wiring for supplying a DC voltage to a load **607**A such as a semiconductor apparatus or the like. The load **607**A is connected to the other end of the output line **622**A. The capacitor **606**A is connected between the output line **622**A and the reference potential **609**A via wirings. In FIG. **7**A, current **612**A illustrated by an arrow represents current occurring in the output line **622**A upstream of the capacitor **606**A.
(51) The DC/DC converter **60**A switches on and off the power, which is supplied from the input voltage source **601**A to the input line **621**A, through switching operations of the switching circuits **602**A, **603**A to feed the power to the downstream inductors **604**A, **605**A. Furthermore, the DC/DC converter **60**A smooths power fed through the inductors **604**A, **605**A at the capacitor **606**A to supply a desired constant voltage to the load **607**A. At this time, the controller **608**A controls timings of the switching operations of the switching circuits **602**A, **603**A. Herein, the switching circuit **602**A and the inductor **604**A form a unit of configuration that functions as a single DC/DC converter. The switching circuit **603**A and the inductor **605**A also form a unit of configuration that functions as a single DC/DC converter. Such units of configuration are each referred to as a

channel. The DC/DC converter **60**A has a first channel formed of the switching circuit **602**A and the inductor **604**A and a second channel formed of the switching circuit **603**A and the inductor **605**A. The switching circuits **602**A, **603**A in respective channels are operated at the same switching frequency. The frequency is several MHz in general.
(52) On the other hand, FIG. **7**B is a circuit diagram illustrating the circuit of the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026. As illustrated in FIG. **7**B, the DC/DC converter **60**B that is the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 has a configuration corresponding to the DC/DC converter **60**A illustrated in FIG. **7**A. In FIG. **7**B, components corresponding to the components illustrated in FIG. **7**A are illustrated with the alphabet of the references being changed from A to B. The DC/DC converter **60**B differs from the DC/DC converter **60**A in that an inductor **604**B has an inductance value L′ that is larger than the inductance value L of the inductor **605**B.
(53) Next, waveforms of the current and the voltage in the DC/DC converters **60**A, **60**B when only a single channel is activated, such as when a semiconductor apparatus that is a load is in a standby state or the like, will be described with reference to FIG. **7**C to FIG. **7**H. FIG. **7**C, FIG. **7**D, and FIG. **7**E illustrate waveforms of the current **610**A, **611**A of the inductors **604**A, **605**A of respective channels, the current **612**A upstream of the capacitor **606**A, and the voltage of the load **607**A in the DC/DC converter **60**A illustrated in FIG. **7**A, respectively. FIG. **7**F, FIG. **7**G, and FIG. **7**H illustrate waveforms of the current **610**B, **611**B of the inductors **604**B, **605**B of respective channels, the current **612**B upstream of the capacitor **606**B, and the voltage of the load **607**B in the DC/DC converter **60**B illustrated in FIG. **7**B, respectively. Herein, for waveform analysis of the current and the voltage, a circuit simulator PSpice by Cadence Design Systems was used. Circuit conditions for the waveform analysis were that the voltages of the input voltage sources **601**A, **601**B were 3 V, the inductance values L, L′ were 0.1 μH, 0.2 μH, respectively, and the switching frequency was 4 MHz. Further, in the waveform analysis, the first channel was operated as the single channel.
(54) As can be seen from FIG. **7**C and FIG. **7**F, in comparison between the current **610**A, **610**B of the first channel, the current **610**B associated with the larger inductance value has a smaller amplitude than the current **610**A. At this time, since only one channel is activated, the current **610**A directly becomes the current **612**A, and the current **610**B directly becomes the current **612**B. In such cases, in the DC/DC converter **60**A, a ripple voltage including ripple noise occurs in the voltage of the load **607**A as a product of the current **612**A and the impedance of the capacitor **606**A. Similarly, in the DC/DC converter **60**B, a ripple voltage including ripple noise occurs in the voltage of the load **607**B as a product of the current **612**B and the impedance of the capacitor **606**B. The amplitude of the ripple voltage in the DC/DC converter **60**B is 16.9 mV as indicated in FIG. **7**H, which means a reduction to approximately a half the amplitude of 33.6 mV of the ripple voltage indicated in FIG. **7**E.
(55) Next, waveforms of the current and the voltage in the DC/DC converters **60**A, **60**B when multiple channels are activated, such as when a semiconductor apparatus is in a full operating state or the like, will be described with reference to FIG. **7**I to FIG. **7**N. FIG. **7**I, FIG. **7**J, and FIG. **7**K illustrate waveforms of the current **610**A, **611**A of the inductors **604**A, **605**A of respective channels, the current **612**A upstream of the capacitor **606**A, and the voltage of the load **607**A in the DC/DC converter **60**A illustrated in FIG. **7**A, respectively. FIG. **7**L, FIG. **7**M, and FIG. **7**N illustrate waveforms of the current **610**B, **611**B of the inductors **604**B, **605**B of respective channels, the current **612**B upstream of the capacitor **606**B, and the voltage of the load **607**B in the DC/DC converter **60**B illustrated in FIG. **7**B, respectively. The waveform analysis of the current and the voltage was performed under the same condition as above except that the first channel and the second channel are activated.
(56) As can be seen from FIG. **7**I, FIG. **7**J, and FIG. **7**K, in the DC/DC converter **60**A that is the general multiphase DC/DC converter illustrated in FIG. **7**A, the current **610**A, **611**A flows with the same amplitude and a phase difference of 180 degrees. Thus, because of the cancelling effect on

both the current **610**A, **611**A, the current **612**A that is a combination thereof has a suppressed, small amplitude as illustrated in FIG. **7**J. As a result, the voltage of the load **607**A has a suppressed, small amplitude as illustrated in FIG. **7**K.

(57) In contrast, as can be seen from FIG. **7**L, in the case of the DC/DC converter **60**B that is the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 illustrated in FIG. **7**B, the amplitudes of the current **610**B, **611**B differ from each other. Thus, the cancelling effect on the current **610**B, **611**B is smaller, and as illustrated in FIG. **7**M, the amplitude of the current **612**B that is a combination of the current **610**B, **611**B is larger. As a result, as illustrated in FIG. **7**N, the ripple voltage appearing at the load **607**B is larger.

(58) As discussed above, in the general multiphase DC/DC converter and the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026, it is difficult to reduce ripple noise in the output voltage regardless of the level of load current. In contrast, DC/DC converters according to first to fourth embodiments of the present disclosure can reduce ripple noise in the output voltage regardless of the level of load current. The DC/DC converter according to each embodiment will be described below. Note that the DC/DC converter according to each embodiment is a multiphase DC/DC converter.

First Embodiment

(59) A DC/DC converter **10** according to the first embodiment of the present disclosure will be described with reference to FIG. **1**A to FIG. **1**D. Note that, in the description, FIG. **7**L and FIG. **7**N illustrating waveforms of the current and the voltage in the DC/DC converter **60**B that is the multiphase DC/DC converter disclosed in Japanese Patent Application Laid-Open No. 2014-226026 illustrated in FIG. **7**B described above will be used for comparison.

(60) First, the configuration of the DC/DC converter **10** according to the present embodiment will be described with reference to FIG. **1**A. FIG. **1**A is a circuit diagram illustrating the circuit of the DC/DC converter **10** according to the present embodiment. The DC/DC converter **10** according to the present embodiment is a two-phase DC/DC converter.

(61) As illustrated in FIG. **1**A, the DC/DC converter **10** according to the present embodiment has switching circuits **102**, **103**, **104**, inductors **105**, **106**, **107**, a capacitor **108**, and a controller **110**. Further, the DC/DC converter **10** has an input line **121** and an output line **122**.

(62) The input line **121** is a wiring supplied with a DC voltage from an input voltage source **101**. The input voltage source **101** is connected to one end of the input line **121**. The input voltage source **101** supplies a DC voltage to the input line **121**. The input voltage source **101** is not particularly limited and may be a power supply that supplies a DC voltage converted from AC power supplied from a commercial power supply or may be a battery, for example. The switching circuits **102**, **103**, **104** are connected in parallel to the other end of the input line **121** via wirings.

(63) The switching circuits **102**, **103**, **104** are each formed of a CMOS inverter having a P-type MOS transistor Tr**1** and an N-type MOS transistor Tr**2**. In each of the switching circuits **102**, **103**, **104**, the source of the P-type MOS transistor Tr**1** is connected to the other end of the input line **121** via a wiring. Further, the source of the N-type MOS transistor Tr**2** is connected to a reference potential **111**, which is the ground potential, via a wiring. Furthermore, the gate of the P-type MOS transistor Tr**1** and the gate of the N-type MOS transistor Tr**2** are connected to the controller **110** via wirings. Further, the drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **102** are connected to each other and connected as the output terminal to one end of the inductor **105** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **103** are connected to each other and connected as the output terminal to one end of the inductor **106** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **104** are connected to each other and connected as the output terminal to one end of the inductor **107** via a wiring. Note that the switching circuits **102**, **103**, **104** are not limited to those formed of CMOS inverters and may take other forms.

(64) The controller **110** is a control unit that controls switching operations of the switching circuits **102**, **103**, **104** to switch on and off the switching circuits **102**, **103**, **104**. The controller **110** controls the switching operation by controlling the voltages supplied to the gates of the P-type MOS transistors Tr**1** and the gates of the N-type MOS transistors Tr**2** of the switching circuits **102**, **103**, **104**.

(65) The inductors **105**, **106**, **107** have inductance values L1, L2, L3, respectively. Herein, the inductance values L1, L2 are larger than the inductance value L3. The inductance values L1, L2 may be the same or may be different from each other. The other end of the inductor **105**, the other end of the inductor **106**, and the other end of the inductor **107** are connected to each other via wirings and connected to one end of the output line **122**. In FIG. **1**A, current **112** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **105** and one end of the output line **122**. Current **113** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **106** and one end of the output line **122**. Current **114** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **107** and one end of the output line **122**.

(66) The output line **122** is a wiring for supplying a DC voltage to a load **109** such as a semiconductor apparatus or the like. The load **109** is connected to the other end of the output line **122**. The capacitor **108** is connected between the output line **122** and the reference potential **111** via wirings. In FIG. **1**A, current **115** illustrated by an arrow represents current occurring in the output line **122** upstream of the capacitor **108**.

(67) The DC/DC converter **10** switches on and off the power, which is supplied from the input voltage source **101** to the input line **121**, through switching operations of the switching circuits **102**, **103**, **104** to feed the power to the downstream inductors **105**, **106**, **107**. Furthermore, the DC/DC converter **10** smooths power fed through the inductors **105**, **106**, **107** at the capacitor **108** to supply a desired constant voltage to the load **109**. At this time, the controller **110** controls timings of the switching operations of the switching circuits **102**, **103**, **104**.

(68) The switching circuit **102** and the inductor **105** form a first channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **103** and the inductor **106** form a second channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **104** and the inductor **107** form a third channel, which is a unit of configuration that functions as a DC/DC converter. The DC/DC converter **10** has three channels of the first channel, the second channel, and the third channel.

(69) In a case of a light-load state, such as when the load **109** is in a standby state or the like, only the first channel formed of the switching circuit **102** and the inductor **105** is activated in the DC/DC converter **10**. In the case of the light-load state, first current flows in the load **109** as the load current in accordance with the state of the load **109**. In the case of the light-load state, the controller **110** controls the switching circuits **102**, **103**, **104** so that only the first channel is activated and the second and third channels are deactivated.

(70) On the other hand, in a case of a heavy-load state where the load is heavier than in the light-load state, such as when the load **109** is in a full operating state or the like, all the channels of the first channel, the second channel, and the third channel are activated in the DC/DC converter **10**. In the case of the heavy-load state, second current that is larger than the first current flows as the load current in the load **109** in accordance with the state of the load **109**. Specifically, in such a case, the first channel formed of the switching circuit **102** and the inductor **105** and the second channel formed of the switching circuit **103** and the inductor **106** are operated in-phase. In contrast, the third channel formed of the switching circuit **104** and the inductor **107** is operated with a phase difference of 180 degrees relative to the first and second channels. In the case of the heavy-load state, the controller **110** controls the switching circuits **102**, **103**, **104** so that the first and second channels are operated in-phase and the third channel is operated with a phase difference of 180 degrees relative to the first and second channels.

(71) It is preferable that the inductance value L1 of the inductor **105** of the first channel, the inductance value L2 of the inductor **106** of the second channel, and the inductance value L3 of the inductor **107** of the third channel satisfy the following Expression (1).

$L3=(L1\times L2)/(L1+L2)$ Expression (1)

(72) When the inductance values L1, L2, L3 satisfy Expression (1), ripple noise can be more effectively reduced due to a cancelling effect between flows of current in a case of a heavy-load state as described later.

(73) Next, the current of each part of the DC/DC converter **10** and the voltage of the load **109** according to the present embodiment illustrated in FIG. **1**A will be described with reference to FIG. **1**B, FIG. **1**C, and FIG. **1**D. FIG. **1**B, FIG. **1**C, and FIG. **1**D illustrate examples of operation waveforms of the DC/DC converter **10** illustrated in FIG. **1**A. FIG. **1**B, FIG. **1**C, and FIG. **1**D illustrate waveforms of the current **112**, **113**, **114** of the inductors **105**, **106**, **107** of respective channels, the current **115** upstream of the capacitor **108**, and the voltage of the load **109** of the DC/DC converter **10**. The waveforms illustrated in FIG. **1**B, FIG. **1**C, and FIG. **1**D are waveforms in a heavy-load state, such as when the load **109** is in a full operating state or the like. Herein, for waveform analysis of the current and the voltage, the circuit simulator PSpice by Cadence Design Systems was used. Circuit conditions for the waveform analysis were that the voltage of the input voltage source **101** was 3 V, the inductance values L1 and L2 of the inductors **105**, **106** each were 0.2 pH, the inductance value L3 of the inductor **107** was 0.1 μH, and the switching frequency was 4 MHz.

(74) In the DC/DC converter **10**, in the heavy-load state, the switching circuits **102**, **103** are operated in synchronization, and the switching circuit **104** is operated with a phase difference of 180 degrees from the switching circuits **102**, **103**.

(75) As illustrated in FIG. **1**, the current **112**, **113** has a waveform having half the amplitude of the current **114** and a phase difference of 180 degrees therefrom. In FIG. **1***i*, the current **112** and the current **113** overlap each other. Therefore, the combined current of the current **112** and the current **113** has a waveform having the same amplitude as the current **114** and a phase difference of 180 degrees therefrom. Thus, due to the cancelling effect between the flows of current, as illustrated in FIG. **1**C, the current **115** has a smaller amplitude than the conventional current **612**B illustrated in FIG. **7**M. As a result, in the present embodiment, ripple noise in the voltage of the load **109** is reduced not only in a case of a light-load state but also in a case of a heavy-load state. While the ripple noise of the conventional DC/DC converter **60**B illustrated in FIG. **7**N is 28.1 mV, the ripple noise is reduced to 23.8 mV in the present embodiment as illustrated in FIG. **1**D. According to the present embodiment, the ripple noise in a heavy-load state can be reduced without impairing the effect of ripple noise reduction in a light-load state compared to the conventional example.

(76) In the present embodiment, since the amplitude of current is suppressed low due to the inductor **105** having a larger inductance value in a case of a light-load state, such as when the load **109** is in a standby state or the like, the ripple noise is reduced. On the other hand, in a case of a heavy-load state where the load is heavier than in the light-load state, such as when the load **109** is in a full operating state or the like, the first and second channels are operated in synchronization. Accordingly, the amplitude of the combined current of the first and second channels is close to or equal to the amplitude of the current of the third channel operated in a different phase from the first and second channel. Thus, in the case of the heavy-load state, ripple noise can be reduced by the cancelling effect between the flows of current. In such a way, according to the present embodiment, ripple noise in the output voltage can be reduced regardless of the level of load current. Note that, although the inductances L1, L2 were the same values in the waveform analysis described above, ripple noise can be reduced even when L1 and L2 are different values from each other as long as both the values are larger than L3. Further, when the value of L1 is larger than the value of L2, ripple noise in a light-load state can be reduced, and it is therefore preferable that the value of L1 be larger than the value of L2.

Second Embodiment

(77) A DC/DC converter **20** according to the second embodiment of the present disclosure will be described with reference to FIG. **2**. FIG. **2** is a circuit diagram illustrating the circuit of the DC/DC converter **20** according to the present embodiment.

(78) As illustrated in FIG. **2**, the DC/DC converter **20** according to the present embodiment has switching circuits **202**, **203**, **204**, inductors **205**, **206**, **207**, **208**, **209**, a capacitor **210**, and a controller **212**. Further, the DC/DC converter **20** has an input line **221** and an output line **222**.

(79) The input line **221** is a wiring supplied with a DC voltage from an input voltage source **201**. The input voltage source **201** is connected to one end of the input line **221**. The input voltage source **201** is the same as the input voltage source **101**. The switching circuits **202**, **203**, **204** are connected in parallel to the other end of the input line **221** via wirings.

(80) The switching circuits **202**, **203**, **204** are each formed of a CMOS inverter in the same manner as the switching circuits **102**, **103**, **104**. In each of the switching circuits **202**, **203**, **204**, the source of the P-type MOS transistor Tr**1** is connected to the other end of the input line **221** via a wiring. Further, the source of the N-type MOS transistor Tr**2** is connected to a reference potential **213**, which is the ground potential, via a wiring. Furthermore, the gate of the P-type MOS transistor Tr**1** and the gate of the N-type MOS transistor Tr**2** are connected to the controller **212** via wirings. Further, the drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **202** are connected to each other and connected as the output terminal to one end of the inductor **205** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **203** are connected to each other and connected as the output terminal to one end of the inductor **207** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **204** are connected to each other and connected as the output terminal to one end of the inductor **209** via a wiring. Note that the switching circuits **202**, **203**, **204** are not limited to those formed of CMOS inverters and may take other forms.

(81) The controller **212** is a control unit that controls switching operations of the switching circuits **202**, **203**, **204** to switch on and off the switching circuits **202**, **203**, **204**. The controller **212** controls the switching operation by controlling the voltages supplied to the gates of the P-type MOS transistors Tr**1** and the gates of the N-type MOS transistors Tr**2** of the switching circuits **202**, **203**, **204**.

(82) The inductors **205**, **206**, **207**, **208**, **209** have the same inductance value L. One end of the inductor **206** is connected to the other end of the inductor **205**. One end of the inductor **208** is connected to the other end of the inductor **207**. The other end of the inductor **206**, the other end of the inductor **208**, and the other end of the inductor **209** are connected to each other via wirings and connected to one end of the output line **222**. In FIG. **2**, current **214** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **206** and one end of the output line **222**. Current **215** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **208** and one end of the output line **222**. Current **216** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **209** and one end of the output line **222**.

(83) The output line **222** is a wiring for supplying a DC voltage to a load **211** such as a semiconductor apparatus or the like. The load **211** is connected to the other end of the output line **222**. The capacitor **210** is connected between the output line **222** and the reference potential **213** via wirings. In FIG. **2**, current **217** illustrated by an arrow represents current occurring in the output line **222** upstream of the capacitor **210**.

(84) The DC/DC converter **20** switches on and off the power, which is supplied from the input voltage source **201** to the input line **221**, through switching operations of the switching circuits **202**, **203**, **204** to feed the power to the downstream inductors **205**, **206**, **207**, **208**, **209**. Furthermore, the DC/DC converter **20** smooths power fed through the inductors **205**, **206**, **207**, **208**, **209** at the

capacitor **210** to supply a desired constant voltage to the load **211**. At this time, the controller **212** controls timings of the switching operations of the switching circuits **202**, **203**, **204**.
(85) The switching circuit **202** and the inductors **205**, **206** form a first channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **203** and the inductors **207**, **208** form a second channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **204** and the inductor **209** form a third channel, which is a unit of configuration that functions as a DC/DC converter. The DC/DC converter **20** has three channels of the first channel, the second channel, and the third channel.
(86) In the DC/DC converter **20** according to the present embodiment, the inductor **105** and the inductor **106** of the DC/DC converter **10** according to the first embodiment illustrated in FIG. **1**A are replaced with the two inductors **205**, **206** and the two inductors **207**, **208**, respectively. In addition, in the present embodiment, the inductors **205**, **206**, **207**, **208**, **209** all have the same inductance value L. For example, when L=0.1 μH, the operation waveforms of the DC/DC converter **20** according to the present embodiment will be the same as the operation waveforms illustrated in FIG. **1**B and FIG. **1**C. That is, also in the present embodiment, the same effect of ripple noise reduction as that in the first embodiment can be obtained.
(87) As described above, in the present embodiment, the inductors **205**, **206** of the first channel are equivalent to two inductors connected in series each being the same as the inductor **209** of the third channel. Further, the inductors **207**, **208** of the second channel are also equivalent to two inductors connected in series each being the same as the inductor **209** of the third channel.
(88) Furthermore, in the present embodiment, when the inductors **205**, **206**, **207**, **208**, **209** are formed of components, there is an advantage that these inductors can be formed of the same components. In general, there are limited lineups of inductor components in which one inductance value is integer multiple of another. In the present embodiment, two inductance components having the same inductance value are connected in series to make a double value, which makes it easier to accurately match the amplitude of the combined current of the first and second channels to the amplitude of the current of the third channel.
(89) Note that the inductors of the first channel and the inductors of the second channel are not limited to two inductors connected in series each being the same as the inductor **209** of the third channel and may be formed of a plurality thereof connected in series, respectively. Also in such a case, the effect of ripple noise reduction can be obtained.
Third Embodiment
(90) Prior to description of a DC/DC converter **40** according to the third embodiment of the present disclosure, a DC/DC converter **30** according to a comparative form to be compared to the DC/DC converter **40** according to the third embodiment will be described with reference to FIG. **3**A to FIG. **3**F. FIG. **3**A is a circuit diagram illustrating the circuit of the DC/DC converter **30** according to the comparative form. The DC/DC converter **30** according to the comparative form is a four-channel multiphase DC/DC converter. FIG. **3**B to FIG. **3**F are diagrams illustrating examples of operation waveforms of the DC/DC converter **30** according to the comparative form.
(91) As illustrated in FIG. **3**A, the DC/DC converter **30** according to the comparative form has switching circuits **302**, **303**, **304**, **305**, inductors **306**, **307**, **308**, **309**, a capacitor **310**, and a controller **312**. Further, the DC/DC converter **30** has an input line **321** and an output line **322**.
(92) The switching circuits **302**, **303**, **304**, **305** are each formed of a CMOS inverter in the same manner as the switching circuits **102**, **103**, **104**. In each of the switching circuits **302**, **303**, **304**, **305**, the source of the P-type MOS transistor Tr**1** is connected to the other end of the input line **321** via a wiring. Further, the source of the N-type MOS transistor Tr**2** is connected to a reference potential **313**, which is the ground potential, via a wiring. Furthermore, the gate of the P-type MOS transistor Tr**1** and the gate of the N-type MOS transistor Tr**2** are connected to the controller **312** via wirings. Further, the drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **302** are connected to each other and connected as the output

terminal to one end of the inductor **306** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **303** are connected to each other and connected as the output terminal to one end of the inductor **307** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **304** are connected to each other and connected as the output terminal to one end of the inductor **308** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **305** are connected to each other and connected as the output terminal to one end of the inductor **309** via a wiring.
(93) The controller **312** is a control unit that controls switching operations of the switching circuits **302**, **303**, **304**, **305** to switch on and off the switching circuits **302**, **303**, **304**, **305**. The controller **312** controls the switching operation by controlling the voltages supplied to the gates of the P-type MOS transistors Tr**1** and the gates of the N-type MOS transistors Tr**2** of the switching circuits **302**, **303**, **304**, **305**.
(94) The inductor **306** has an inductance value L′. The inductors **307**, **308**, **309** each have an inductance value L that is smaller than the inductance value L′. The other end of the inductor **306**, the other end of the inductor **307**, the other end of the inductor **308**, and the other end of the inductor **309** are connected to each other via wirings and connected to one end of the output line **322**. In FIG. **3**A, current **314** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **306** and one end of the output line **322**. Current **315** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **307** and one end of the output line **322**. Current **316** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **308** and one end of the output line **322**. Current **317** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **309** and one end of the output line **322**.
(95) The output line **322** is a wiring for supplying a DC voltage to a load **311** such as a semiconductor apparatus or the like. The load **311** is connected to the other end of the output line **322**. The capacitor **310** is connected between the output line **322** and the reference potential **313** via wirings. In FIG. **3**A, current **318** illustrated by an arrow represents current occurring in the output line **322** upstream of the capacitor **310**.
(96) The DC/DC converter **30** switches on and off the power, which is supplied from the input voltage source **301** to the input line **321**, through switching operations of the switching circuits **302**, **303**, **304**, **305** to feed the power to the downstream inductors **306**, **307**, **308**, **309**. Furthermore, the DC/DC converter **30** smooths power fed through the inductors **306**, **307**, **308**, **309**, at the capacitor **310** to supply a desired constant voltage to the load **311**. At this time, the controller **312** controls timings of the switching operations of the switching circuits **302**, **303**, **304**, **305**.
(97) The switching circuit **302** and the inductor **306** form a first channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **303** and the inductor **307** form a second channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **304** and the inductor **308** form a third channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **305** and the inductor **309** form a fourth channel, which is a unit of configuration that functions as a DC/DC converter. The DC/DC converter **30** has four channels of the first channel, the second channel, the third channel, and the fourth channel.
(98) In a case of a light-load state, such as when the load **311** is in a standby state or the like, only the first channel formed of the switching circuit **302** and the inductor **306** having the larger inductance value L′ is activated in the DC/DC converter **30**. At this time, the remaining channels are deactivated. In the case of the light-load state, first current flows in the load **311** as the load current in accordance with the state of the load **311**. In the case of the light-load state, the controller **312** controls the switching circuits **302**, **303**, **304**, **305** so that only the first channel is activated and the second, third, and fourth channels are deactivated.

(99) On the other hand, in a case of a heavy-load state, such as when the load **311** is in a full operating state, all the channels of the first, second, third, and fourth channels are activated in the DC/DC converter **30**. At this time, however, respective channels are operated with a phase difference of 90 degrees from each other. In the case of the heavy-load state, second current that is larger than the first current flows in the load **311** as the load current in accordance with the state of the load **311**. In the case of the heavy-load state, the controller **312** controls the switching circuits **302**, **303**, **304**, **305** so that the first, second, third, and fourth channels are operated with a phase shift by 90 degrees from the each other.
(100) Next, the current of each part of the DC/DC converter **30** and the voltage of the load **311** according to the comparative form illustrated in FIG. **3**A will be described with reference to FIG. **3**B, FIG. **3**C, FIG. **3**D, FIG. **3**E, and FIG. **3**F. FIG. **3**B, FIG. **3**C, FIG. **3**D, FIG. **3**E, and FIG. **3**F illustrate examples of operation waveforms of the DC/DC converter **30** illustrated in FIG. **3**A. Herein, for waveform analysis of the current and the voltage, the circuit simulator PSpice by Cadence Design Systems was used. Circuit conditions for the waveform analysis were that the voltage of the input voltage source **301** was 3 V, the inductance value L′ of the inductor **306** was 0.2 μH, the inductance value L of the inductors **307**, **308**, **309** was 0.1 μH, and the switching frequency was 4 MHz.
(101) FIG. **3**B and FIG. **3**C illustrate the waveforms of the current **318** and the voltage of the load **311** when only the first channel is activated in a light-load state, respectively. FIG. **3**D, FIG. **3**E, and FIG. **3**F illustrate the waveforms of the current **314**, **315**, **316**, **317**, current **318**, and the voltage of the load **311** when all the four channels are activated in a heavy-load state, respectively.
(102) As can be seen from FIG. **3**D, in the heavy-load state, although the phase difference between respective current **314**, **315**, **316**, **317** is 90 degrees, only the current **314** of the inductor **306** having a larger inductance value than the remaining inductors **307**, **308**, **309** has a smaller amplitude. Thus, as illustrated in FIG. **3**E, the current **318**, which is the combined current that is a combination of the current **314**, **315**, **316**, **317**, has a waveform with a small cancelling effect. As a result, as illustrated in FIG. **3**F, the ripple noise in the voltage of the load **311** was 22.1 mV.
(103) Next, the DC/DC converter **40** according to the third embodiment of the present disclosure will be described with reference to FIG. **4**A to FIG. **4**F. FIG. **4**A is a circuit diagram illustrating the circuit of the DC/DC converter **40** according to the present embodiment. FIG. **4**B to FIG. **4**F are diagrams illustrating examples of operation waveforms of the DC/DC converter **40** according to the present embodiment.
(104) As illustrated in FIG. **4**A, the DC/DC converter **40** according to the present embodiment has switching circuits **402**, **403**, **404**, **405**, **406**, inductors **407**, **408**, **409**, **410**, **411**, a capacitor **412**, and a controller **414**. Further, the DC/DC converter **40** has an input line **431** and an output line **432**.
(105) The input line **431** is a wiring supplied with a DC voltage from an input voltage source **401**. The input voltage source **401** is connected to one end of the input line **431**. The input voltage source **401** is the same as the input voltage source **101**. The switching circuits **402**, **403**, **404**, **405**, **406** are connected in parallel to the other end of the input line **431** via wirings.
(106) The switching circuits **402**, **403**, **404**, **405**, **406** are each formed of a CMOS inverter in the same manner as the switching circuits **102**, **103**, **104**. In each of the switching circuits **402**, **403**, **404**, **405**, **406**, the source of the P-type MOS transistor Tr is connected to the other end of the input line **431** via a wiring. Further, the source of the N-type MOS transistor Tr**2** is connected to a reference potential **415**, which is the ground potential, via a wiring. Furthermore, the gate of the P-type MOS transistor Tr**1** and the gate of the N-type MOS transistor Tr**2** are connected to the controller **414** via wirings. Further, the drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **402** are connected to each other and connected as the output terminal to one end of the inductor **407** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **403** are connected to each other and connected as the output terminal to one end of the inductor **408** via a

wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **404** are connected as the output terminal to each other and connected to one end of the inductor **409** via a wiring. The drain of the P-type MOS transistor Tr**1** and the drain of the N-type MOS transistor Tr**2** of the switching circuit **405** are connected to each other and connected as the output terminal to one end of the inductor **410** via a wiring. The drain of the P-type MOS transistor Tr and the drain of the N-type MOS transistor Tr**2** of the switching circuit **406** are connected to each other and connected as the output terminal to one end of the inductor **411** via a wiring. Note that the switching circuits **402**, **403**, **404**, **405**, **406** are not limited to those formed of CMOS inverters and may take other forms.

(107) The controller **414** is a control unit that controls switching operations of the switching circuits **402**, **403**, **404**, **405**, **406** to switch on and off the switching circuits **402**, **403**, **404**, **405**, **406**. The controller **414** controls the switching operation by controlling the voltages supplied to the gates of the P-type MOS transistors Tr and the gates of the N-type MOS transistors Tr**2** of the switching circuits **402**, **403**, **404**, **405**, **406**.

(108) The inductor **407** has an inductance value L1. The inductor **408** has an inductance value L2. The inductors **409**, **410**, **411** have the same inductance value L3. The inductance value L3 is a smaller than the inductance values L1, L2. The inductance values L1, L2 may be the same or may be different from each other. The other end of the inductor **407**, the other end of the inductor **408**, the other end of the inductor **409**, the other end of the inductor **410**, and the other end of the inductor **411** are connected to each other via wirings and connected to one end of the output line **432**. In FIG. **4**A, current **416** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **407** and one end of the output line **432**. Current **417** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **408** and one end of the output line **432**. Current **418** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **409** and one end of the output line **432**. Current **419** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **410** and one end of the output line **432**. Current **420** illustrated by an arrow represents current occurring in a wiring connected between the other end of the inductor **411** and one end of the output line **432**.

(109) The output line **432** is a wiring for supplying a DC voltage to a load **413** such as a semiconductor apparatus or the like. The load **413** is connected to the other end of the output line **432**. The capacitor **412** is connected between the output line **432** and the reference potential **415** via wirings. In FIG. **4**A, current **421** illustrated by an arrow represents current occurring in the output line **432** upstream of the capacitor **412**.

(110) The DC/DC converter **40** switches on and off the power, which is supplied from the input voltage source **401** to the input line **431**, through switching operations of the switching circuits **402**, **403**, **404**, **405**, **406** to feed the power to the downstream inductors **407**, **408**, **409**, **410**, **411**. Furthermore, the DC/DC converter **40** smooths power fed through the inductors **407**, **408**, **409**, **410**, **411** at the capacitor **412** to supply a desired constant voltage to the load **413**. At this time, the controller **414** controls timings of the switching operations of the switching circuits **402**, **403**, **404**, **405**, **406**.

(111) The switching circuit **402** and the inductor **407** form a first channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **403** and the inductor **408** form a second channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **404** and the inductor **409** form a third channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **405** and the inductor **410** form a fourth channel, which is a unit of configuration that functions as a DC/DC converter. The switching circuit **406** and the inductor **411** form a fifth channel, which is a unit of configuration that functions as a DC/DC converter. The DC/DC converter **40** has five channels of the first channel, the second channel, the third channel, the fourth channel, and the fifth channel.

(112) In a case of a light-load state, such as when the load **413** is in a standby state or the like, only the first channel formed of the switching circuit **402** and the inductor **407** having the larger inductance value L1 is activated in the DC/DC converter **40**. At this time, the remaining channels are deactivated. In the case of the light-load state, first current flows in the load **413** as the load current in accordance with the state of the load **413**. In the case of the light-load state, the controller **414** controls the switching circuits **402**, **403**, **404**, **405**, **406** so that only the first channel is activated and the second, third, fourth, and fifth channels are deactivated.
(113) On the other hand, in a case of a heavy-load state, such as when the load **413** is in a full operating state, all the channels of the first, second, third, fourth, and fifth channels are activated in the DC/DC converter **40**. At this time, however, the first channel and the second channel, which is formed of the switching circuit **403** and the inductor **408** having the larger inductance value L2, are operated in-phase. On the other hand, the remaining third to fifth channels are operated with a phase shift by 90 degrees each from the phase in which the first and second channels are operated. In the case of the heavy-load state, the second current that is larger than the first current flows in the load **413** as the load current in accordance with the state of the load **413**. In the case of the heavy-load state, the controller **414** controls the switching circuits **402**, **403** so that the first and second channels are operated in-phase. Furthermore, in such a case, the controller **414** controls the switching circuits **404**, **405**, **406** so that the third, fourth, and fifth channels are operated with a phase shift by 90 degrees each from the phase in which the first and second channels are operated.
(114) It is preferable that the inductance value L1 of the inductor **407** of the first channel, the inductance value L2 of the inductor **408** of the second channel, and the inductance value L3 of the inductors **409**, **410**, **411** of the third to fifth channels satisfy the following Expression (2).
$L3=(L1\times L2)/(L1+L2)$ Expression (2)
(115) When the inductance values L1, L2, L3 satisfy Expression (2), ripple noise can be more effectively reduced due to a cancelling effect between flows of current in a case of a heavy-load state as described later.
(116) Next, the current of each part of the DC/DC converter **40** and the voltage of the load **413** according to the present embodiment illustrated in FIG. **4**A will be described with reference to FIG. **4**B, FIG. **4**C, FIG. **4**D, FIG. **4**E, and FIG. **4**F. FIG. **4**B, FIG. **4**C, FIG. **4**D, FIG. **4**E, and FIG. **4**F illustrate examples of operation waveforms of the DC/DC converter **40** illustrated in FIG. **4**A. Herein, for waveform analysis of the current and the voltage, the circuit simulator PSpice by Cadence Design Systems was used. Circuit conditions for the waveform analysis were that the voltage of the input voltage source **401** was 3 V, the inductance values L1, L2 of the inductors **407**, **408** each were 0.2 μH, the inductance value L3 of the inductors **409**, **410**, **411** was 0.1 μH, and the switching frequency was 4 MHz.
(117) FIG. **4**B and FIG. **4**C illustrate waveforms of the current **421** and the voltage of the load **413** when only the first channel is activated in a light-load state, respectively. FIG. **4**D, FIG. **4**E, and FIG. **4**F illustrate waveforms of the current **416**, **417**, **418**, **419**, **420**, the current **421**, and the voltage of the load **413** when all the five channels are activated in a heavy-load state, respectively.
(118) When all the five channels are activated in the heavy-load state, such as when the load **413** is in a full operating state, the current **416**, **417** has waveforms having half the amplitude of the current **418**, **419**, **420**, as illustrated in FIG. **4**D. In FIG. **4**D, the current **416** and the current **417** overlap each other. Therefore, the combined current of the current **416** and the current **417** has the same amplitude as the current **418**, **419**, **420**, and the ripple noise is reduced due to the cancelling effect between the flows of current. While the ripple noise of the DC/DC converter **30** according to the comparative form illustrated in FIG. **3**F is 22.1 mV, the ripple noise is reduced to 13 mV in the DC/DC converter **40** according to the present embodiment as illustrated in FIG. **4**F.
(119) Note that, when only the first channel is activated in a light-load state, the amplitude of ripple noise is 17.1 mV as illustrated in FIG. **4**C in the DC/DC converter **40** according to the present embodiment. This result is the same as the result illustrated in FIG. **3**C of the DC/DC converter **30**

according to the comparative form.

(120) As described above, according to the present embodiment, ripple noise in the output voltage can be reduced regardless of the level of load current.

(121) Note that, although the case where the DC/DC converter **40** has the fourth and fifth channels including the inductors **410**, **411** having the same inductance value as the inductor **409** of the third channel has been described in the present embodiment, the invention is not limited thereto. The DC/DC converter **40** can have N (N being an integer greater than or equal to one) channels including inductors having the same inductance value as the inductor **409** of the third channel, respectively. The configurations of the N channels are the same as the fourth channel, respectively. In such a case, the controller **414** controls switching circuits of the first, second, and third channels and the N channels. Accordingly, in a case of a heavy-load state, the controller **414** controls the switching circuit of each channel so that the third channel and the N channels are operated with a phase shift by 360/(N+2) degrees each from the phase in which the first and second channels are operated.

(122) Further, also in the present embodiment, the inductor of the first channel and the inductor of the second channel may be formed of a plurality of inductors connected in series each being the same as the inductor of the third channel in the same manner as in the second embodiment.

Fourth Embodiment

(123) A DC/DC converter according to the fourth embodiment of the present disclosure will be described with reference to FIG. **5**A to FIG. **5**E. FIG. **5**A to FIG. **5**E are diagrams illustrating operation waveforms of the DC/DC converter according to the present embodiment.

(124) The configuration of the DC/DC converter according to the present embodiment is the same as the configuration of the DC/DC converter **40** according to the third embodiment illustrated in FIG. **4**A. In the present embodiment, description will be provided for a case where circuit conditions differ from those of the fourth embodiment.

(125) FIG. **5**A, FIG. **5**B, FIG. **5**C, FIG. **5**D, and FIG. **5**E are diagrams illustrating operation waveforms when the circuit conditions of the DC/DC converter **40** illustrated in FIG. **4**A are set as follows. Herein, for waveform analysis of the current and the voltage, the circuit simulator PSpice by Cadence Design Systems was used. Circuit conditions for the waveform analysis were that the voltage of the input voltage source **401** was 3 V, the inductance value L1 of the inductor **407** was 0.47 μH, the inductance value L2 of the inductor **408** was 0.13 μH, and the inductance value L3 of the inductors **409**, **410**, **411** was 0.1 μH. The switching frequency was 4 MHz.

(126) As described above, in the present embodiment, the inductor **407** of the first channel has a larger inductance value than the inductor **408** of the second channel.

(127) FIG. **5**A and FIG. **5**B illustrate waveforms of the current **421** and the voltage of the load **413** when only the first channel is activated in a light-load state, respectively. In the present embodiment, due to an effect of the increase of the inductance value L1 of the inductor of the first channel from 0.2 μH to 0.47 μH, the amplitude of ripple noise is reduced from 17.1 mV to 7.4 mV as illustrated in FIG. **5**B compared to the third embodiment illustrated in FIG. **4**B and FIG. **4**C.

(128) On the other hand, FIG. **5**C, FIG. **5**D, and FIG. **5**E illustrate the current **416**, **417**, **418**, **419**, **420**, the current **421**, and the voltage of the load **413** when all the five channels are activated in a heavy-load state, respectively. As illustrated in FIG. **5**C, while the current **416** and the current **417** have the same phase and different amplitudes, the sum of both the amplitudes is equal to the amplitude of the current **418**, **419**, **420**. Therefore, the combined current of the current **416**, **417** and the remaining current **418**, **419**, **420** have the same amplitude with a phase difference of 90 degrees, respectively. Thus, due to the cancelling effect between the flows of current, the current **421** that is the combined current of all the current **416**, **417**, **418**, **419**, **420** is reduced as illustrated in FIG. **5**D. As a result, in the present embodiment, the amplitude of ripple noise is 12.4 mV as illustrated in FIG. **5**E, which is substantially the same as the amplitude of ripple noise of 13 mV illustrated in FIG. **4**F of the third embodiment.

(129) As described above, in the present embodiment, the ripple noise in a light-load state can be further reduced compared to the third embodiment.
(130) Note that, also in the present embodiment, the inductor of the first channel and the inductor of the second channel may be formed of a plurality of inductors connected in series each being the same as the inductor of the third channel in the same manner as in the second embodiment.
Fifth Embodiment
(131) An electronic apparatus according to the fifth embodiment of the present disclosure will be described with reference to FIG. **6**A to FIG. **6**C. FIG. **6**A is a sectional view illustrating a digital camera **500** that is an imaging apparatus as an example of the electronic apparatus according to the present embodiment. FIG. **6**B and FIG. **6**C are a perspective view and a sectional view illustrating a processing module **504** included in the digital camera **500**, respectively. In the present embodiment, the digital camera **500** that is an electronic apparatus including any one of the DC/DC converters **10**, **20**, **40** according to the first to fourth embodiments described above will be described.
(132) As illustrated in FIG. **6**A, the digital camera **500** that is the imaging apparatus as an example of the electronic apparatus according to the present embodiment is an interchangeable lens digital camera, for example, and has a camera body **501**. A lens barrel (lens unit) **502** including lenses is removably attached to the camera body **501**. Note that the digital camera **500** is not limited to the interchangeable lens type and may be an integrated lens type, for example.
(133) The camera body **501** has a casing **503**, the processing module **504** that is a printed circuit board, and a sensor module **505** that is a printed circuit board. The processing module **504** and the sensor module **505** are arranged inside the casing **503**. The processing module **504** and the sensor module **505** are electrically connected to each other via the cable **506**. The processing module **504** and the sensor module **505** are examples of a semiconductor module that is an electronic module.
(134) The sensor module **505** has an image sensor **5051** that is an image pickup device and a printed wiring board **5052**. The image sensor **5051** is mounted and implemented on the printed wiring board **5052**. For example, the image sensor **5051** is a complementary metal oxide semiconductor (CMOS) image sensor or a charge coupled device (CCD) image sensor. The image sensor **5051** has a function of converting light incident via the lens unit **502** into an electrical signal.
(135) As illustrated in FIG. **6**B and FIG. **6**C, the processing module **504** has a semiconductor apparatus **5041** that is, for example, an application specific integrated circuit (ASIC), a power source circuit **5042**, and a printed wiring board **5043**. The semiconductor apparatus **5041** and the power source circuit **5042** are mounted and implemented on the printed wiring board **5043**. For example, the printed wiring board **5043** is a rigid substrate, which is a member on which the semiconductor apparatus **5041** and the power source circuit **5042** are mounted. The semiconductor apparatus **5041** is a digital signal processor, for example, and has a function of acquiring an electrical signal from the image sensor **5051** and performing a process of correcting the acquired electrical signal to generate image data. The power source circuit **5042** is formed of any one of the DC/DC converters **10**, **20**, **40** according to the first to fourth embodiments. The power source circuit **5042** converts a DC voltage supplied from a battery (not illustrated) into a predetermined voltage and supplies the predetermined voltage to the semiconductor apparatus **5041**.
(136) Note that, although the digital camera **500** has been described as the electronic apparatus in the present embodiment, the invention is not limited thereto. Electronic apparatuses including any one of the DC/DC converters **10**, **20**, **40** may be any electronic apparatuses other than a digital camera.
(137) According to various embodiments of the present disclosure, ripple noise in the output voltage can be reduced regardless of the level of load current.
(138) While various embodiments of the present disclosure have been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed

exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(139) This application claims the benefit of Japanese Patent Application No. 2022-089697, filed Jun. 1, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A DC/DC converter comprising: an input line supplied with a DC voltage; a plurality of channels each including a switching circuit and an inductor, one end of the inductor being connected to the input line via the switching circuit; an output line, one end of the output line being connected to the other end of the inductors of the plurality of channels, and the other end of the output line being connected to a load; and a controller that controls the switching circuits of the plurality of channels, wherein the plurality of channels include at least a first channel, a second channel, and a third channel, wherein the inductors of the first and second channels each have a larger inductance value than an inductance value of the inductor of the third channel, wherein in a case where a current flowing in the load is a first current, the controller controls the switching circuits of the first to third channels so that only the first channel is activated and the second and third channels are deactivated, and wherein in a case where the current flowing in the load is a second current that is larger than the first current, the controller controls the switching circuits of the first to third channels so that the first and second channels are operated in-phase and the third channel is operated in a phase different from the phase of the first and second channels.
2. The DC/DC converter according to claim 1, wherein in a case where the current flowing in the load is the second current that is larger than the first current, the controller controls the switching circuits of the first to third channels so that the third channel is operated with a phase difference of **180** degrees relative to the first and second channel.
3. The DC/DC converter according to claim 1, wherein the plurality of channels further includes N channels, where N is an integer greater than or equal to one, and wherein each inductor of the N channels has an inductance value that is the same as the inductance value of the inductor of the third channel.
4. The DC/DC converter according to claim 3, wherein in a case where the current flowing in the load is the second current that is larger than the first current, the controller controls the switching circuits of the first to third channels and switching circuits of the N channels so that the first channel and the second channel are operated in-phase and the third channel and the N channels are operated with a phase shift by 360/(N+2) degrees each from the phase in which the first and second channels are operated.
5. The DC/DC converter according to claim 3, wherein the inductor of the first channel has a larger inductance value than an inductance value of the inductor of the second channel.
6. The DC/DC converter according to claim 1, wherein the inductors of the first and second channels are each formed of a plurality of inductors connected in series each being the same as the inductor of the third channel.
7. An electronic module comprising: a member; and a DC/DC converter according to claim 1, the DC/DC converter being mounted on the member.
8. An electronic apparatus comprising: a casing; and an electronic module arranged inside the casing, wherein the electronic module comprises: a member; and a DC/DC converter according to claim 1, the DC/DC converter being mounted on the member.
9. A DC/DC converter comprising: an input line supplied with a DC voltage; a plurality of channels each including a switching circuit and an inductor, one end of the inductor being connected to the input line via the switching circuit; an output line, one end of the output line being connected to the other end of the inductors of the plurality of channels, and the other end of the output line being connected to a load; and a controller that controls the switching circuits of the plurality of channels,

wherein the plurality of channels include at least a first channel, a second channel, and a third channel, wherein the inductors of the first and second channels each have a larger inductance value than an inductance value L**3** of the inductor of the third channel, wherein in a case where a current flowing in the load is a first current, the controller controls the switching circuits of the first to third channels so that only the first channel is activated and the second and third channels are deactivated, wherein in a case where the current flowing in the load is a second current that is larger than the first current, the controller controls the switching circuits of the first to third channels so that the first and second channels are operated in-phase and the third channel is operated in a phase different from the phase of the first and second channels, and wherein an inductance value L1 of the inductor of the first channel, an inductance value L2 of the inductor of the second channel, and the inductance value L3 of the inductor of the third channel satisfy following Expression (1):

$$L3=(L1\times L2)/(L1+L2) \qquad (1).$$

10. The DC/DC converter according to claim 9, wherein in a case where the current flowing in the load is the second current that is larger than the first current, the controller controls the switching circuits of the first to third channels so that the third channel is operated with a phase difference of 180 degrees relative to the first and second channels.

11. The DC/DC converter according to claim 9, wherein the plurality of channels further includes N channels, where N is an integer greater than or equal to one, and wherein each inductor of the N channels has an inductance value that is the same as the inductance value L3 of the inductor of the third channel.

12. The DC/DC converter according to claim 11, wherein, in a case where the current flowing in the load is the second current that is larger than the first current, the controller controls the switching circuits of the first to third channels and switching circuits of the N channels so that the first channel and the second channel are operated in-phase and the third channel and the N channels are operated with a phase shift by 360/(N+2) degrees each from the phase in which the first and second channels are operated.

13. The DC/DC converter according to claim 11, wherein the inductance value L1of the inductor of the first channel is larger than the inductance value L2 of the inductor of the second channel.

14. The DC/DC converter according to claim 9, wherein the inductors of the first and second channels are each formed of a plurality of inductors connected in series each being the same as the inductor of the third channel.

15. An electronic module comprising: a member; and a DC/DC converter according to claim 9, the DC/DC converter being mounted on the member.

16. An electronic apparatus comprising: a casing; and an electronic module arranged inside the casing, wherein the electronic module comprises: a member; and a DC/DC converter according to claim 9, the DC/DC converter being mounted on the member.

17. A DC/DC converter comprising: an input line supplied with a DC voltage; a plurality of channels each including a switching circuit and an inductor, one end of the inductor being connected to the input line via the switching circuit; an output line, one end of the output line being connected to the other end of the inductors of the plurality of channels, and the other end of the output line being connected to a load; and a controller that controls the switching circuits of the plurality of channels, wherein the plurality of channels include at least a first channel, a second channel, and a third channel, wherein the inductors of the first and second channels each have a larger inductance value than an inductance value L**3** of the inductor of the third channel, wherein in a case where a current flowing in the load is a first current, the controller controls the switching circuits of the first to third channels so that only the first channel is activated and the second and third channels are deactivated, wherein in a case where the current flowing in the load is a second current that is larger than the first current, the controller controls the switching circuits of the first to third channels so that the first and second channels are operated in-phase and the third channel is

operated in a phase different from the phase of the first and second channels, wherein the plurality of channels further includes N channels, where N is an integer greater than or equal to one, wherein each inductor of the N channels has an inductance value that is the same as the inductance value L3 of the inductor of the third channel, and wherein an inductance value L1 of the inductor of the first channel, an inductance value L2 of the inductor of the second channel, and the inductance value L3 of the inductor of the third channel satisfy following Expression (2):

$L3=(L1\times L2)/(L1+L2) \qquad (2).$

18. The DC/DC converter according to claim 17, wherein in a case where the current flowing in the load is the second current that is larger than the first current, the controller controls the switching circuits of the first to third channels so that the third channel is operated with a phase difference of 180 degrees relative to the first and second channels.

19. The DC/DC converter according to claim 17, wherein, in a case where the current flowing in the load is the second current that is larger than the first current, the controller controls the switching circuits of the first to third channels and switching circuits of the N channels so that the first channel and the second channel are operated in-phase and the third channel and the N channels are operated with a phase shift by 360/(N+2) degrees each from the phase in which the first and second channels are operated.

20. The DC/DC converter according to claim 17, wherein the inductance value L1of the inductor of the first channel is larger than the inductance value L2 of the inductor of the second channel.

21. The DC/DC converter according to claim 17, wherein the inductors of the first and second channels are each formed of a plurality of inductors connected in series each being the same as the inductor of the third channel.

22. An electronic module comprising: a member; and a DC/DC converter according to claim 19, the DC/DC converter being mounted on the member.

23. An electronic apparatus comprising: a casing; and an electronic module arranged inside the casing, wherein the electronic module comprises: a member; and a DC/DC converter according to claim 17, the DC/DC converter being mounted on the member.